

Title: AI-Powered Virtual Interview and Candidate Assessment Platform Using Speech and NLP.

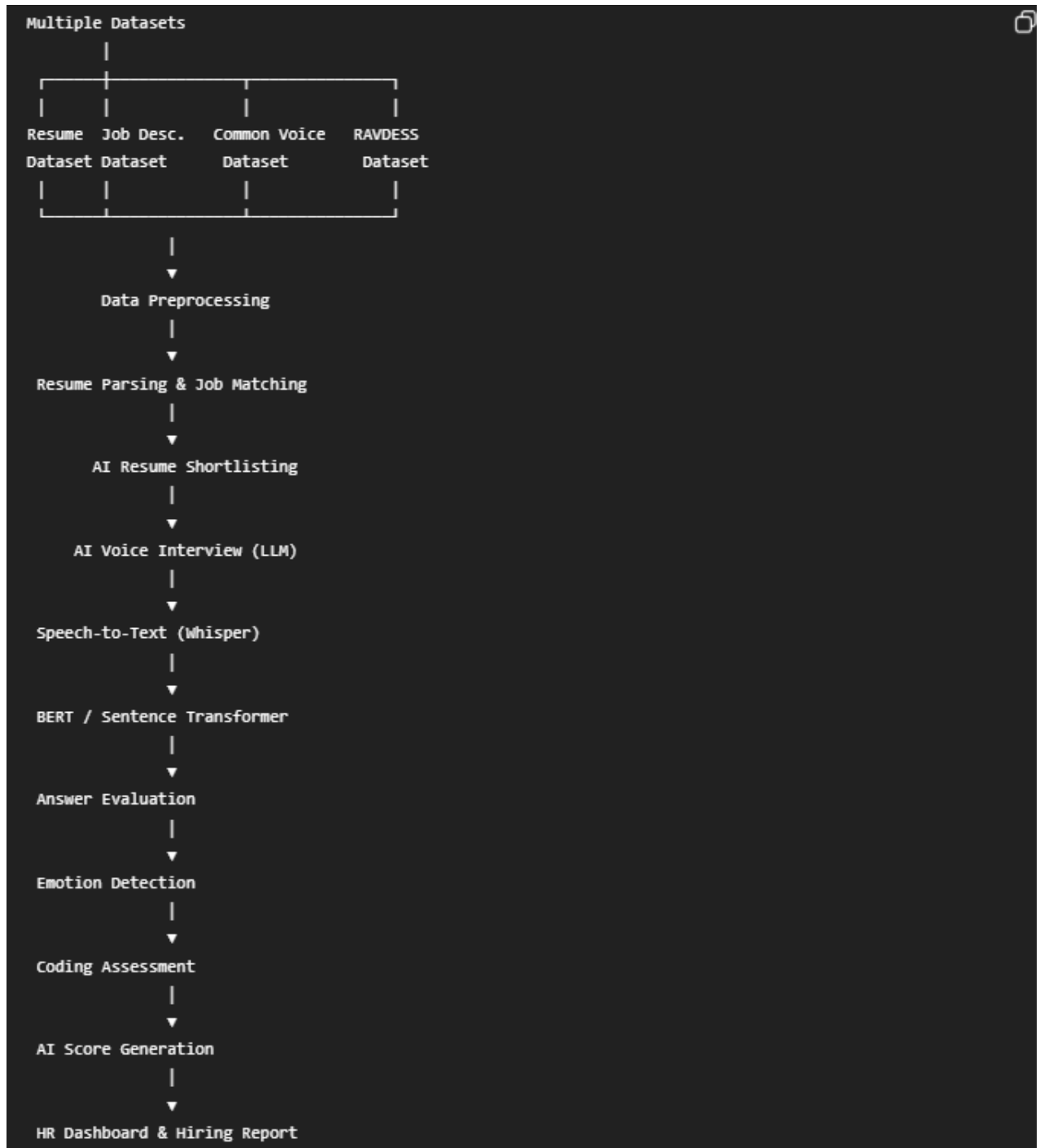
Abstract:

Recruitment is a time-consuming process for organizations due to manual resume screening and interviews. This project proposes an **AI-Powered Virtual Interview and Candidate Assessment Platform** that automates the initial hiring process using Artificial Intelligence, Natural Language Processing (NLP), and Speech Processing. The system first analyses the candidate's resume and compares it with the job description to perform resume shortlisting. Shortlisted candidates attend an AI-based voice interview where the AI asks domain-specific technical questions based on the candidate's resume and job requirements. Candidate responses are converted into text using the Whisper Speech-to-Text model and evaluated using Transformer-based language models such as BERT and Sentence Transformers. The system also performs speech emotion analysis and communication assessment to estimate confidence and speaking quality. A short domain-specific coding assessment is conducted at the end of the interview. Finally, the platform generates an AI-based evaluation report containing resume matching score, technical performance, communication score, coding score, confidence analysis, and an overall hiring recommendation for HR. The proposed system aims to provide a faster, fairer, and more intelligent recruitment process.

Datasets:

Dataset	URL Link
Resume Dataset	https://www.kaggle.com/datasets/snehaanbhawal/resume-dataset
Job Description Dataset	https://www.kaggle.com/datasets/ravindrasinghrana/job-description-dataset
Mozilla Common Voice Dataset	https://mozilladatacollective.com/organization/cmfmh0j9o10006ns07jq45h7xk
RAVDESS Emotional Speech Dataset	https://www.kaggle.com/datasets/uwrkagglerravdess-emotional-speech-audio
CodeSearchNet Dataset	https://github.com/github/CodeSearchNet

Approach:



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